

Emergency Startup & Pressure Control Protocol for FCC High-Pressure Separator (V-204)

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Facility	Area 200 - Hydrocracker & Catalytic Cracking Complex (Jamnagar Refinery Unit)
Effective Date	July 15, 2024
Compliance	OISD-STD-117, PESO Static Pressure Vessels Rules 2016, ISO 45001:2018

Purpose

This SOP defines emergency startup, pressure control, Lockout/Tagout (LOTO), emergency depressurization, and Root Cause Analysis guidance for FCC High-Pressure Separator V-204.

Equipment Specifications

Parameter	Value
Primary Asset	V-204 High-Pressure Gas/Liquid Separator
Connected Pump	P-101A Crude Charge Pump A
MAWP	52.0 bar
Normal Operating Pressure	45.2 bar
Operating Temperature	380°C
High Pressure Trip	49.5 bar
PSV-204A Lift Setpoint	51.0 bar

Emergency Startup Procedure

- 1. Verify mechanical readiness, PTW, instrument calibration, PSV certification, and EDV functionality.
- 2. Perform four-step nitrogen purge: (a) pressurize to 2–3 barg with 99.99% N2 for 10 min; (b) vent to flare until O2<2% and HC<10% LEL; (c) repeat purge cycle three times; (d) maintain 0.2–0.5 barg positive pressure.
- 3. Start auxiliary systems, cooling water, lubrication, then Pump P-101A. Introduce feed gradually while maintaining pressure near 45.2 bar.
- 4. Continuously monitor PT-204A/B, separator level, pump vibration, and seal leakage. Respond immediately above 48.5 bar.

LOTO Procedure

- Stop Pump P-101A and isolate electrical supply.
- Close Valve V-204-IN-01, apply lock and danger tag.
- Close Valve V-204-OUT-02, apply lock and tag.
- Depressurize, drain, vent, and verify zero energy before maintenance.

Emergency Depressurization

Actuate EDV-204 for pressure above 49.5 bar, fire exposure, vessel damage, or major process upset. Confirm flare availability and depressurize in a controlled manner.

Root Cause Analysis

Incident ID: INC-2024-88

Event Date: May 12, 2024

Failure Summary: Mechanical seal failure on Pump P-101A caused a secondary pressure spike in Separator V-204.

Root Cause: Thermal stress cracking of silicon carbide seal faces caused by transient cooling water flow interruption due to CW-201 line throttling.

CAPA: Replace seal assembly, install dual magnetic face bearing isolators, deploy continuous DP acoustic sensor monitoring, recalibrate cooling water controls, update startup checklist, and enhance predictive maintenance.